

Identify corresponding, alternate and co-interior relationships between angles formed when parallel lines are crossed by a transversal; use them to solve problems and explain reasons

Learning intention

- identify corresponding, alternate and co-interior relationships between angles formed when parallel lines are crossed by a transversal; use them to solve problems and explain reasons.

Teach and model

- using geometric reasoning of angle properties to generalise the angle relationships of parallel lines and transversals.
- constructing a pair of parallel lines and a pair of perpendicular lines using their properties, a pair of compasses and a ruler.
- using dynamic geometry software to demonstrate how angles and their properties are involved in the design and construction of scissor lifts, folding umbrellas.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

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Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.